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# INTRODUCTION

## What is DSMAC?

- DSMAC (Digital Scene Matching Area Correlation) is a vision-based navigation technique that estimates a vehicle's position by matching onboard sensor imagery with reference maps.
- It is widely used in GPS-denied environments where satellite navigation is unavailable or unreliable.
- DSMAC compares observed scene features with stored terrain or scene information to determine accurate localization.

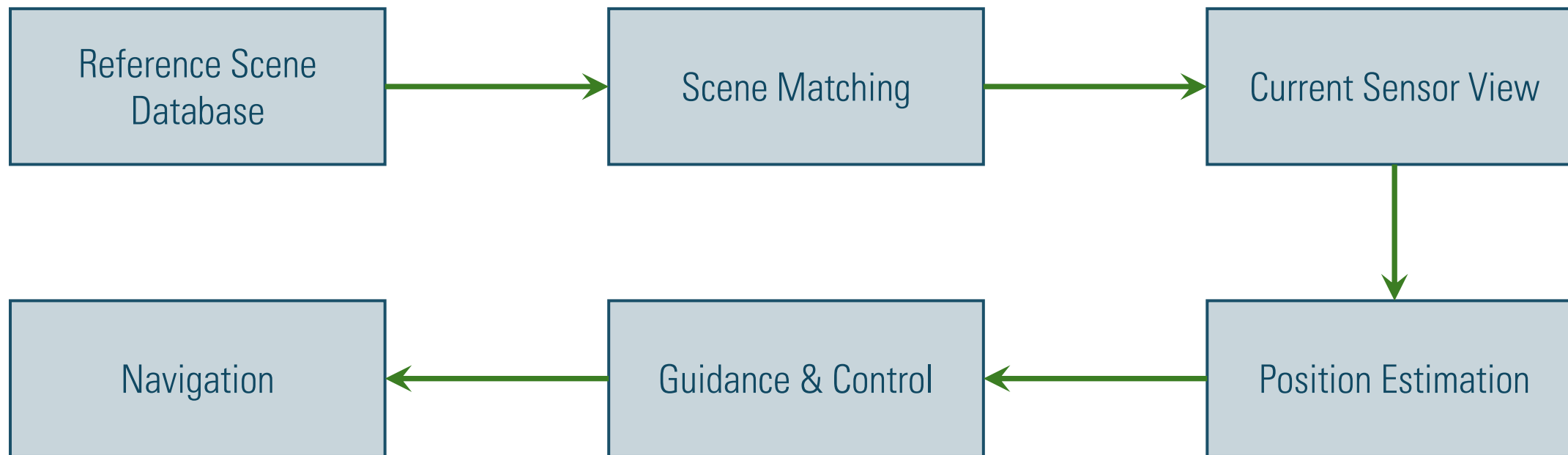
## Why is DSMAC Important?

- Provides autonomous navigation without dependence on GPS.
- Enhances robustness in contested or signal-denied environments.
- Applicable to aerial vehicles, missiles, autonomous robots, and unmanned systems.
- Enables precise localization using onboard vision sensors.

## Motivation

Modern navigation systems increasingly employ multiple sensing modalities, such as RGB and infrared (IR) cameras, to improve robustness under varying environmental and illumination conditions.

# DSMAC NAVIGATION FRAMEWORK

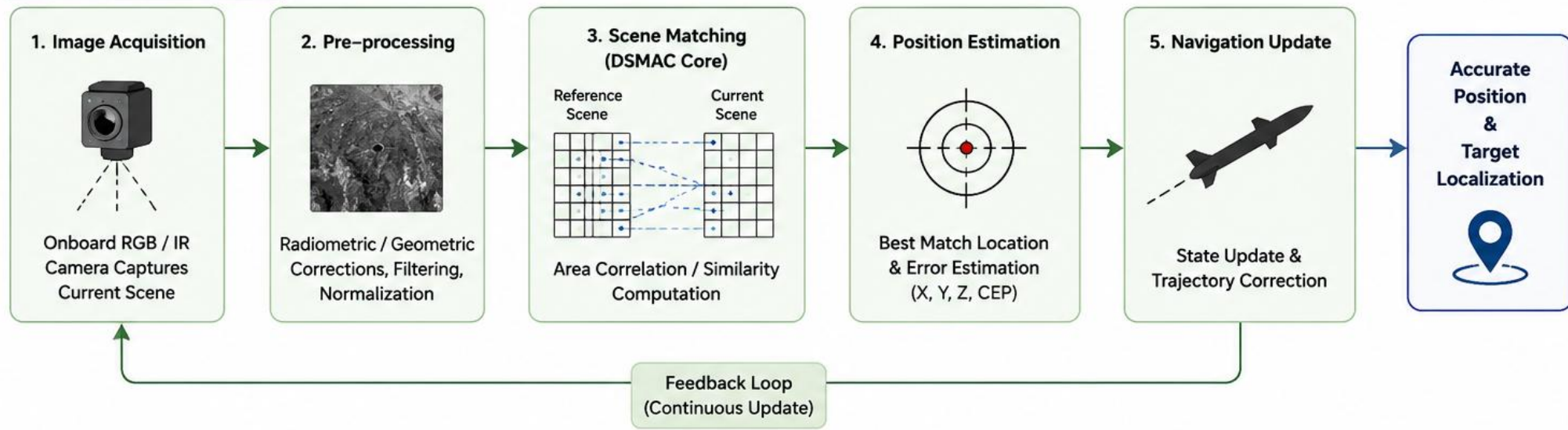


# DSMAC – GENERAL WORKING PIPELINE

## PRE-MISSION (OFFLINE)



## IN-MISSION (ONLINE)



# LITERATURE REVIEW

## 1. Original DSMAC (1980)

### Reference

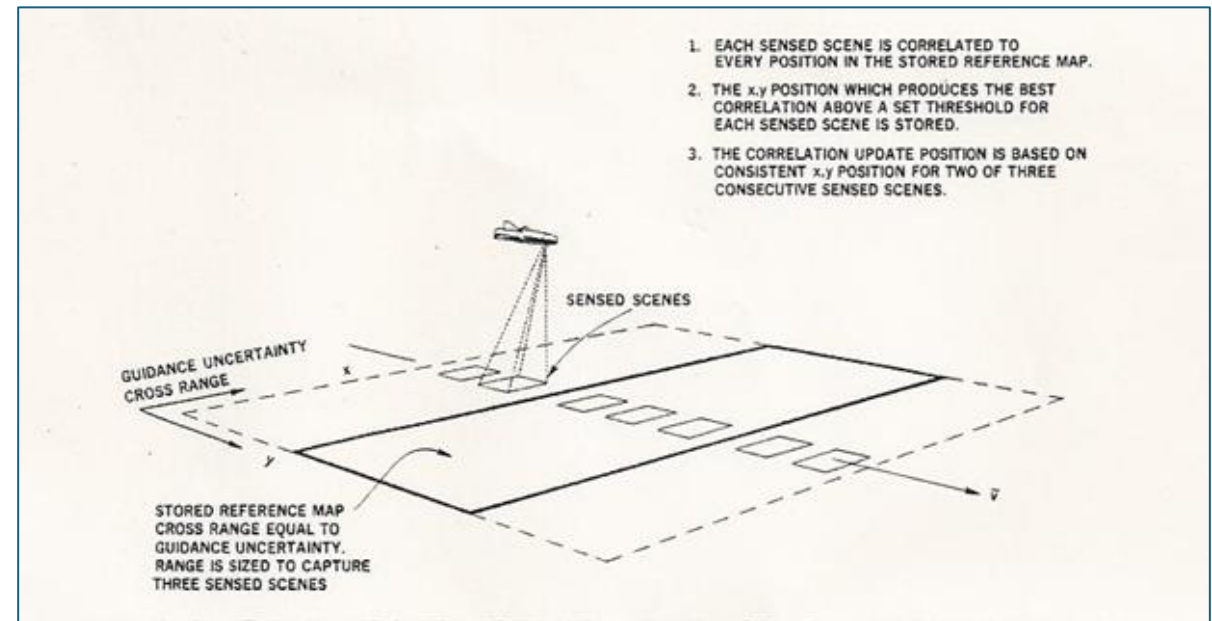
- Carr & Sobek, *Digital Scene Matching Area Correlator (DSMAC)*, Naval Avionics Center, 1980

### Approach

- Area-based image correlation
- Correlation Surface Peak Detection
- Pixel-wise scene comparison
- DSMAC-assisted terminal guidance
- Reference image matching

### Limitations

- Sensitive to illumination and viewpoint changes
- High computational complexity
- Limited adaptability



# LITERATURE REVIEW

## 2. Advanced Scene Matching (1994)

### Reference

- Irani & Christ, *Image Processing for Tomahawk Scene Matching*, Johns Hopkins APL Technical Digest, 1994

### Approach

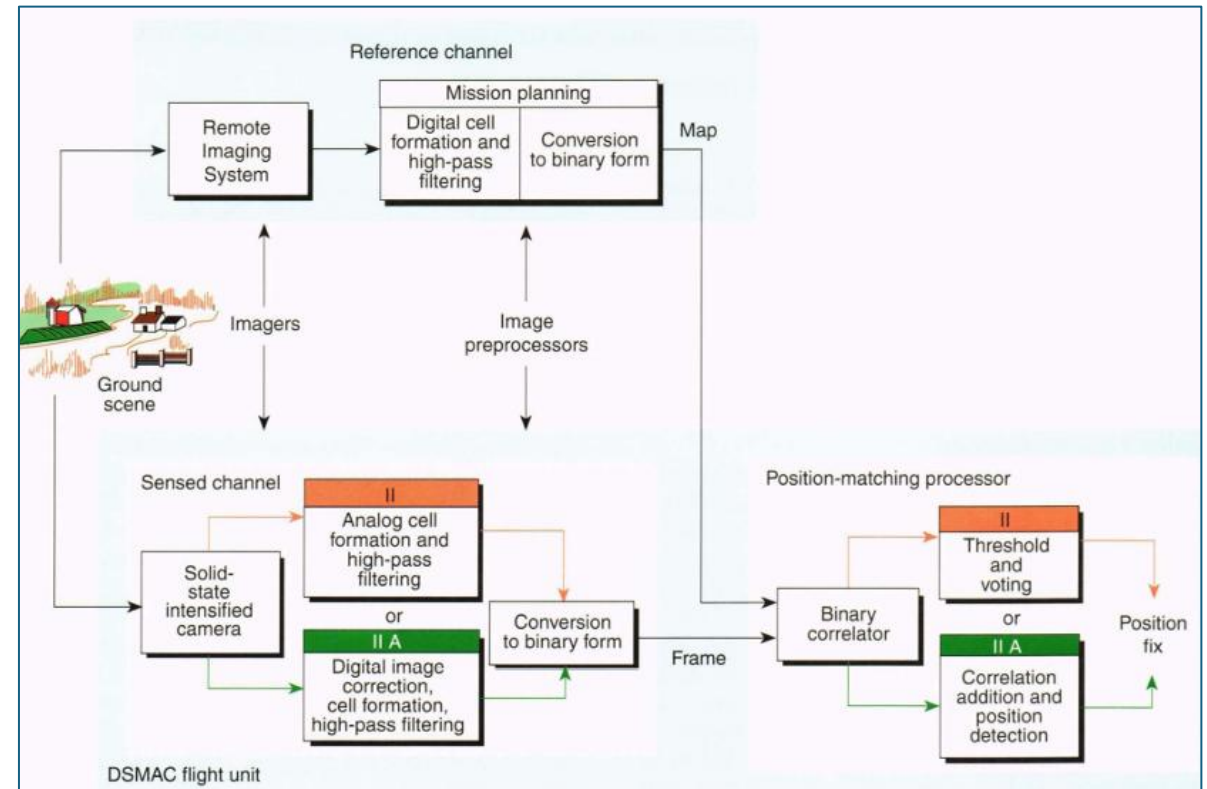
- Enhanced image processing techniques
- Improved scene registration
- Robust geometric alignment
- Optimized image matching for terminal guidance

### Contributions

- Improved robustness over classical DSMAC
- Better handling of scene distortions
- Increased localization accuracy

### Limitations

- Still relied on handcrafted image processing
- No learned feature representation
- Sensitive to environmental variations



# LITERATURE REVIEW

## 3. AI-Based DSMAC (2026)

### Reference

- Kashyap et al., *ML Based Solution for DSMAC System in Aerial Navigation*, 2026.
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### Approach

- ORB feature extraction
- ResNet-50 feature learning
- FAISS similarity search
- CNN-based aerial image matching

### Contributions

- Deep feature representation
- Faster large-scale image retrieval
- Improved robustness for aerial navigation
- Automated dataset generation

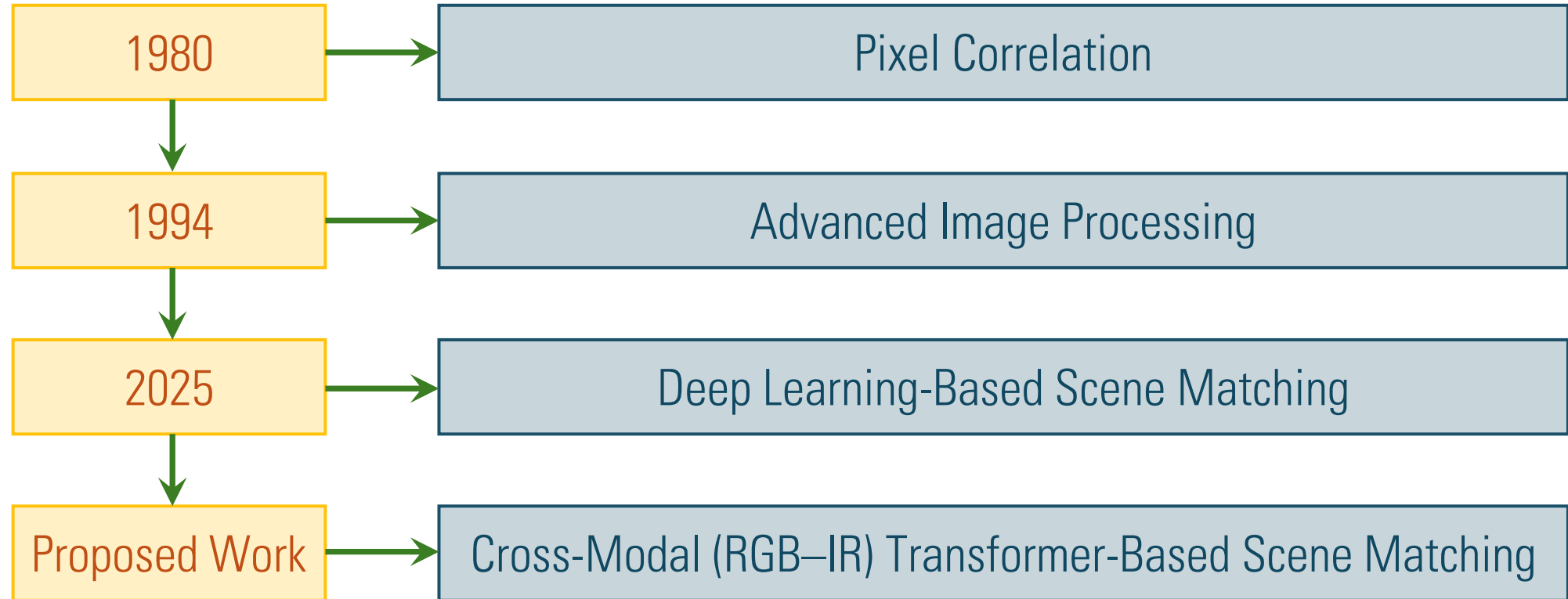
### Limitations

- High computational requirements
- Limited multimodal (RGB–IR) capability

Table 5. Model Comparison Overview

| Model | Accuracy | Processing Time | Strengths                      | Limitations                     |
|-------|----------|-----------------|--------------------------------|---------------------------------|
| ORB   | High     | 15 s            | Accurate matching of keypoints | Too slow for real-time use      |
| FAISS | Moderate | less than 1 s   | Efficient, robust and scalable | Required pre-trained embeddings |

# EVOLUTION OF DSMAC IMAGE MATCHING



# CURRENT STATE-OF-THE-ART FEATURE MATCHING METHODS

| Method                       | Approach                                           | Strength                                     | Limitation                                |
|------------------------------|----------------------------------------------------|----------------------------------------------|-------------------------------------------|
| Superpoint<br>+<br>Superglue | Sparse keypoints + GNN<br>matching                 | Accurate and robust<br>matching              | High computational cost;<br>RGB-trained   |
| LoFTR                        | Detector-free transformer-<br>based dense matching | Excellent in low-texture<br>regions          | Computationally intensive;<br>RGB-focused |
| Efficient LoFTR              | Lightweight transformer-<br>based dense matching   | Faster inference with<br>comparable accuracy | Limited validation on RGB-<br>IR imagery  |

## CURRENT STATE-OF-THE-ART FEATURE MATCHING METHODS (CONTD.)

| Method      | Approach                                | Strength                            | Limitation                                              |
|-------------|-----------------------------------------|-------------------------------------|---------------------------------------------------------|
| MatchFormer | Coarse-to-fine hierarchical transformer | Robust multi-scale feature matching | High memory usage; not optimized for real-time          |
| ASpanFormer | Adaptive span attention transformer     | Handles large viewpoint changes     | Complex architecture; higher inference latency          |
| RoMa        | Regression-based dense correspondence   | Accurate dense pixel-wise matching  | Computationally expensive; limited real-time deployment |
| LightGlue   | Adaptive transformer feature matcher    | Fast and efficient feature matching | Sparse matching; depends on extracted keypoints         |

# COMPARATIVE ANALYSIS OF EXISTING IMAGE MATCHING METHODS

| Method                 | Matching Strategy  | Real-Time Performance | RGB–IR Support | Major Limitation                                |
|------------------------|--------------------|-----------------------|----------------|-------------------------------------------------|
| Classical DSMAC (1980) | Area Correlation   | Low                   | No             | Sensitive to illumination and viewpoint changes |
| Advanced DSMAC (1994)  | Scene Registration | Moderate              | No             | Handcrafted processing, limited adaptability    |
| ORB                    | Keypoint Matching  | High                  | No             | Performance degrades in feature-poor regions    |
| SuperPoint             | Keypoint Detection | High                  | No             | Trained primarily on RGB datasets               |

# COMPARATIVE ANALYSIS OF EXISTING IMAGE MATCHING METHODS (CONTD.)

| Method          | Matching Strategy                | Real-Time Performance | RGB–IR Support | Major Limitation                          |
|-----------------|----------------------------------|-----------------------|----------------|-------------------------------------------|
| SuperGlue       | Graph Neural Network`            | Moderate              | No             | High computational complexity             |
| LoFTR           | Transformer-based Dense Matching | Moderate              | No             | Computationally intensive                 |
| EfficientLoFTER | Efficient Transformer            | High                  | No             | Limited evaluation on cross-modal imagery |
| LightGlue       | Adaptive Transformer             | High                  | No             | Sparse correspondence, RGB-focused        |

# LIMITATIONS OF EXISTING APPROACHES

## Classical DSMAC

- Relies on pixel-level correlation, making it highly sensitive to illumination and viewpoint variations.
- Performance degrades significantly under dynamic environmental conditions.
- Computationally expensive for large search spaces.

## Feature-Based Methods

- Depend on handcrafted feature descriptors.
- Reduced matching accuracy in texture-poor and repetitive environments.
- Limited robustness against large appearance changes.

## Deep Learning-Based Methods

- Primarily trained and evaluated on RGB image datasets.
- Most methods assume same-modality image pairs during feature matching.
- Computational complexity limits deployment in high-frequency navigation systems.
- Limited evaluation under real-time embedded constraints.

# RESEARCH GAP

Lack of **modality-invariant feature representations** for RGB–IR scene matching.

Absence of robust cross-modal correspondence estimation in existing DSMAC systems.

Need for dense feature matching with low computational latency.

Existing methods do not simultaneously satisfy **accuracy, robustness, and real-time performance (50–100 Hz)** required for next-generation DSMAC.

# PROPOSED METHODOLOGY

## Objective

To develop a **real-time multimodal DSMAC framework** capable of robust scene matching between RGB and infrared imagery by learning a shared latent feature space and performing efficient transformer-based feature correspondence.

## Proposed Framework

### Step 1: Multimodal Image Acquisition

- Simultaneous acquisition of RGB and Infrared (IR) images using synchronized onboard sensors.
- Image preprocessing including geometric alignment and normalization.

### Step 2: Dual Encoder Network

- Independent feature encoders for RGB and IR images.
- Learn modality-specific representations while preserving structural scene information.
- Project both modalities into a **shared latent feature space**.

# PROPOSED METHODOLOGY (CONTD.)

## Step 3: Shared Latent Space Learning

- Align RGB and IR feature embeddings using a shared feature representation.
- Reduce modality discrepancy while preserving semantic correspondence.
- Generate modality-invariant descriptors suitable for feature matching.

## Step 4: EfficientLoFTR-based Feature Matching

- Apply EfficientLoFTR to perform dense feature correspondence directly in the shared latent space.
- Utilize transformer-based attention mechanisms for robust matching.
- Eliminate dependency on handcrafted keypoint detectors.

## Step 5: DSMAC Localization

- Estimate scene correspondences between the onboard image and reference database.
- Compute localization error.
- Update navigation state and trajectory correction.
- Repeat continuously for real-time guidance.

# PROPOSED METHODOLOGY (CONTD.)

## Expected Advantages

Robust RGB–IR feature correspondence.

Modality-invariant feature representation.

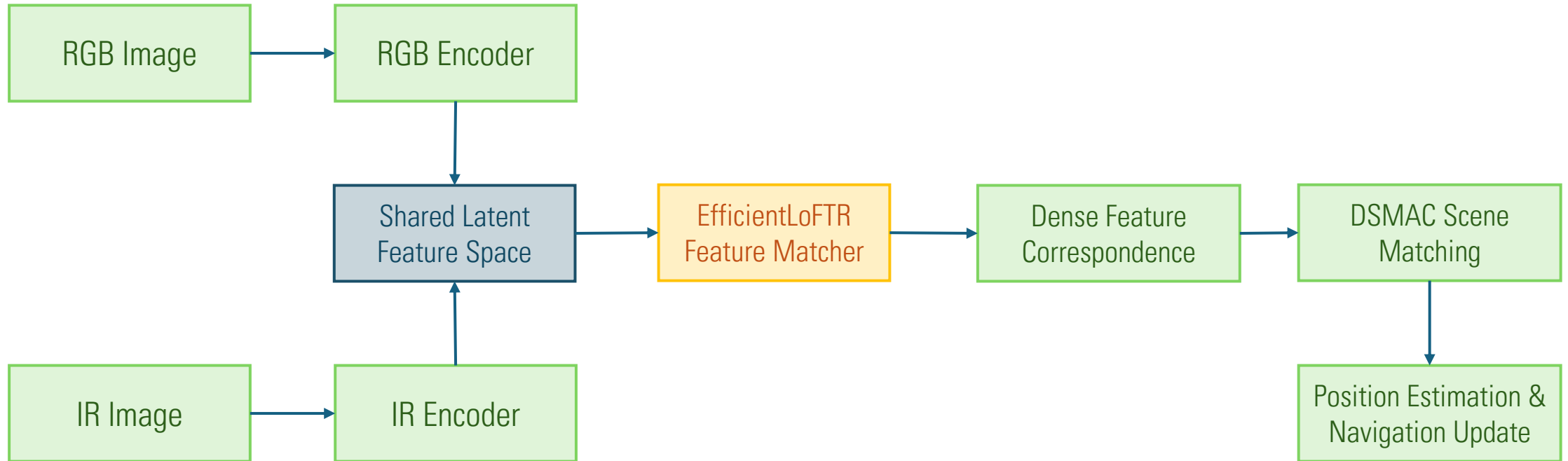
Reduced computational overhead using EfficientLoFTR.

Target real-time performance of **50–100 Hz**.

Improved localization accuracy under varying illumination conditions.

Scalable architecture for integration with next-generation autonomous navigation systems.

## PROPOSED METHODOLOGY (CONTD.)



# REFERENCES

- Carr & Sobek, **Digital Scene Matching Area Correlator (DSMAC)**, *Naval Avionics Center*, 1980
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